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Sustainable Industrial Wastewater Treatment: Assessing Green Chemistry-Driven In-Situ Oxidant Technology

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Abstract

Industrial wastewater treatment often faces challenges, particularly when contaminant loads vary and upset processes occur. This study explores a system designed to generate sodium ferrate-based oxidants in situ through an innovative, renewables-integrated electrolysis process. These oxidants are dosed directly into industrial wastewater streams. Laboratory evaluations were conducted across multiple industrial water samples to assess the performance of this oxidant technology under varying contamination profiles.

Wastewater composition varies significantly across industrial applications, making standardized treatment approaches difficult to implement effectively. The proposed system demonstrates promising results compared to conventional treatment methods, particularly in handling high-load and variable-contaminant streams. The system's modular design enhances adaptability and reduces carbon footprint by enabling on-site oxidant generation.

The process eliminates the need for chlorine-based or de-chlorination techniques, instead employing environmentally friendly oxidants capable of degrading persistent and complex contaminants, including heavy metals. Sodium ferrate, generated in situ as the primary oxidant, reacts with contaminants to form stable, low-volume sediments. Moreover, its high oxidation potential enables effective breakdown of pollutants without the logistical challenges of transporting hazardous materials. This study highlights how integrated, site-specific oxidant generation can offer a sustainable, scalable solution for industrial wastewater treatment.

Introduction

The membrane-based electrolysis cell developed in this study can generate sodium ferrate for industrial wastewater treatment, offering an environmentally sustainable and cost-effective solution. Its modular and adaptive design can simplify deployment across a range of industrial scenarios [Ref 5]. The system uses an alkali solution to produce sodium ferrate, demonstrating improved oxidation efficiency compared to other electrolytes, such as potassium-based systems [Ref 1,2]. The use of consumable anodes accelerates decomposition reactions within the cell, enhancing ferrate generation and overall process yield.

Sodium ferrate can be applied in smaller quantities than many conventional treatment agents, owing to its strong oxidation potential. It decomposes complex contaminants into non-toxic end products, generating environmentally benign sediments. This minimizes the risk to aquatic life and reduces the secondary pollution burden posed by other oxidants. Ferrates also break down toxic organic compounds and mitigate microbial risks [Ref 6].

The electrolysis cell's configuration allows for continuous in situ ferrate generation, which enhances treatment responsiveness and eliminates the need for hazardous chemical transport. This also improves the stability and effectiveness of the treatment by avoiding ferrate decomposition prior to use [Ref 4]. The process shall include the estimation of the emissions for natural or forced ventilation based on the volume of the room and the volumes of the gases or vapors. For ferrates, less than 200 grams per day could be handled with natural ventilation and production within the range of 200 grams to 1 Kg shall be monitored with gas detectors and production above 1 kg shall be administered with sensors and forced ventilation [Ref 6].

Industrial Relevance and Research Motivation

Industrial growth is directly linked to increasing demand for raw materials, water, and energy. As water is consumed across various production processes, it becomes contaminated with a wide range of pollutants—from process chemicals to heavy metals—making effective and adaptable treatment systems essential. Each industry generates wastewater of differing composition, depending on the application and operational complexity.

Wastewater from petrochemical plants, for example, is often rich in Total Organic Compounds (TOC), Volatile Organic Compounds (VOC), Biological Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Dissolved Solids (TDS), Total Suspended Solids (TSS), and heavy metals. These pollutants arise due to extensive contact with raw materials, catalysts, solvents, and crude derivatives during processing. Similarly, fertilizer manufacturing contributes streams with high ammonia concentrations, while metalworking and coatings industries frequently discharge water containing heavy metals and TDS.

The diversity of contaminant profiles poses a challenge to conventional treatment systems, which are often designed around predictable influent quality. With no universally accepted method to treat these highly variable streams, there is a need to develop modular, adaptable technologies capable of on-site customization.

This study investigates the in-situ generation and application of sodium ferrate—a powerful oxidant—as a sustainable and adaptable treatment solution. The performance of this technology is evaluated across diversified industrial wastewater sources, focusing on its ability to reduce key contamination parameters.

Applicable Standards and Regulations

Numerous local and international regulatory frameworks govern industrial wastewater discharge to ensure environmental safety. In Saudi Arabia, these include:

1. Royal Commission Environmental Regulations 2015
2. Saudi Aramco Engineering Standard SAES-A-103
3. Ministry of Environment, Water And Agriculture

These standards define allowable limits for contaminants and outline discharge conditions to the marine environment, groundwater, or for re-use. This study aligns with these benchmarks in evaluating the practical applicability of sodium ferrate treatment across varying industrial scenarios.

Problem Statement Guiding Research Questions

Industrial wastewater treatment faces two persistent challenges: (1) the *increasing variability and complexity of contaminant loads* across different industrial sectors, and (2) the *lack of adaptable, environmentally sustainable treatment systems* that can respond to these variations in real time. Existing centralized systems are often ill-equipped to manage high concentrations of recalcitrant pollutants such as heavy metals, VOCs, and TDS, especially in sectors like petrochemicals, fertilizers, and metal finishing. Additionally, treatment approaches that rely on the transportation and storage of chemical oxidants present logistical, economic, and environmental risks. There is a critical need for modular, on-site treatment technologies that can generate high-potency oxidants sustainably and adapt to diverse wastewater profiles.

1. Can sodium ferrate be effectively generated on-site through a membrane-based electrolysis system?
2. How does the in-situ generated sodium ferrate perform across different categories of industrial wastewater (e.g., petrochemical, fertilizer, metal finishing)?
3. How does the process compare in terms of treatment efficiency, environmental impact, and operational feasibility against traditional oxidant-based methods?
4. What are the sedimentation characteristics, handling implications, and downstream environmental risks associated with ferrate use versus other reagents?

Principle of Operation and Cell Design

Figure 1A illustrates a schematic of the membrane-based electrolysis cell used for the in-situ generation of sodium ferrate. In this system, cations (e.g., Na^+) migrate through an ion exchange membrane from the cathode chamber toward the anode, while the entire system is maintained in an alkaline environment to facilitate ferrate formation.

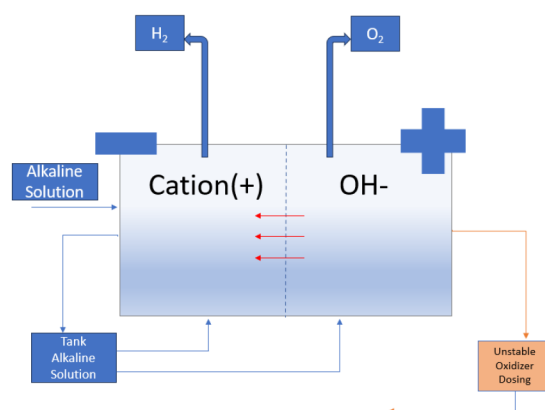


Figure 1A—Schematic of Electrolysis Cell for In-Situ Sodium Ferrate Generation

The cell is designed as a divided electrolyzer, with separate compartments for the anode and cathode. This configuration prevents the mixing of evolved gases—primarily hydrogen at the cathode and oxygen at the anode—thereby eliminating the risk of explosive gas mixtures and reducing the need for inert gas purging. The ion exchange membrane ensures ion selectivity, enhances operational safety, and improves energy efficiency by minimizing cross-reactions.

Sodium ferrate (Na_2FeO_4) is generated at the anode via oxidation of iron in the presence of sodium hydroxide. The oxidant is dosed directly into the target wastewater stream without intermediate storage, improving both stability and environmental performance. Unlike undivided cells, this design avoids gas recombination and reduces parasitic losses [Ref 3].

Estimates from international studies suggest that the capital and operating costs for this system are lower than those for alternative oxidant generation methods such as in-situ hypochlorite, ozone, or ultraviolet disinfection systems.

Methodology and Process Overview

As illustrated in Figure 1B, the process consists of three stages:

- Stage 1: Sodium ferrate is generated in a membrane-based electrolysis cell using a sodium hydroxide solution.
- Stage 2: The oxidant is dosed into the contaminated wastewater for reaction.
- Stage 3: The treated water undergoes filtration to remove suspended solids and reaction by-products.

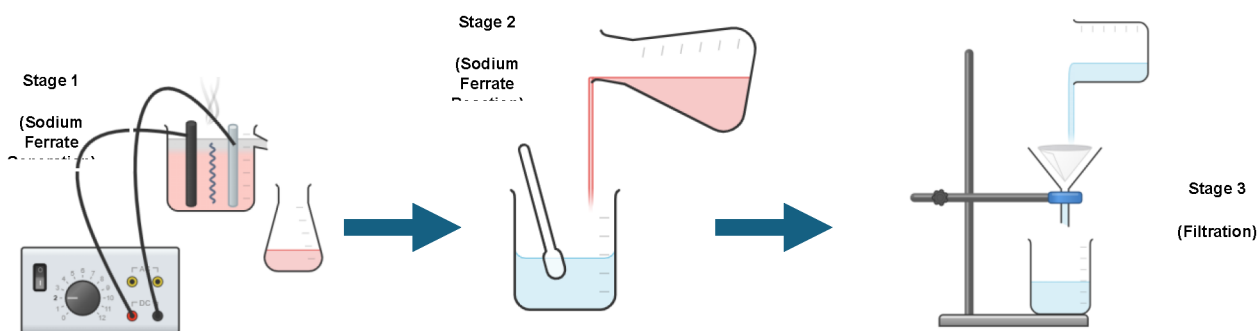


Figure 1B—Principle Process Steps for wastewater treatment through sodium ferrate

A compact, lab-scale prototype (Figure 2) was developed with dedicated anode and cathode chambers separated by an ion exchange membrane. The reactor includes a ferrate collection chamber and utilizes ferrate-resistant materials, such as polypropylene-reinforced glass fiber. A concentration sensor regulates the dosage by controlling reagent production in real time, based on ferrate concentration.

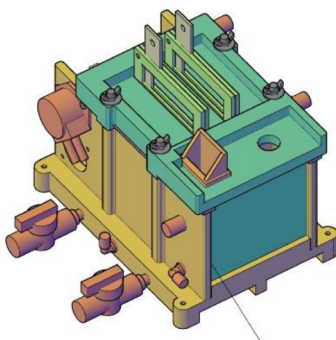


Figure 2—3D Model of Sodium Ferrate generator

Electric current is applied across the electrodes, enabling oxidation at the anode and hydrogen evolution at the cathode. The ion exchange membrane restricts ion crossover and gas mixing, enhancing safety and efficiency. At typical flow rates, less than 200 grams per day of sodium ferrate can be produced using ≤ 5 A current. For larger-scale applications, gas monitoring and controlled ventilation are recommended to manage evolved gases and avoid hazardous accumulation.

Process Control and Automation

The dosage rates and recirculation rates could be controlled by software as well for automated processes and the process could be run in manual mode as well. To automate the process for the complex stream sodium ferrate generator and dosing of sodium ferrate could be equipped with controller to adjust the power output for the generator to control the production and concentration of the sodium ferrate. Figure 2 shows the sodium ferrate generator 3D model equipped with anode, cathode, ion exchange membrane, pumps and supply drain valves.

Treatment Process Flow

The full treatment sequence is depicted in Figures 3A and 3B. Sodium ferrate generated in the electrolyzer is introduced into the wastewater stream in a reaction column, where oxidation occurs. The water then flows into a settling tank, allowing particulate matter and coagulated compounds to settle. Subsequent filtration stages remove residual solids of various sizes, ensuring compliance with discharge standards.

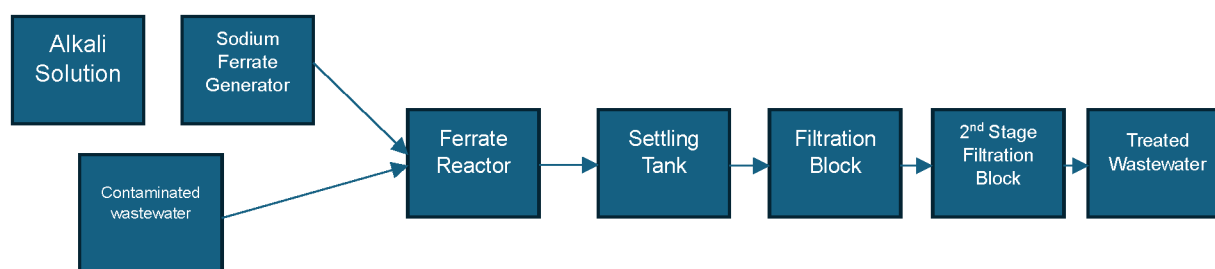


Figure 3A—Schematic For Sodium Ferrate Reaction Stages

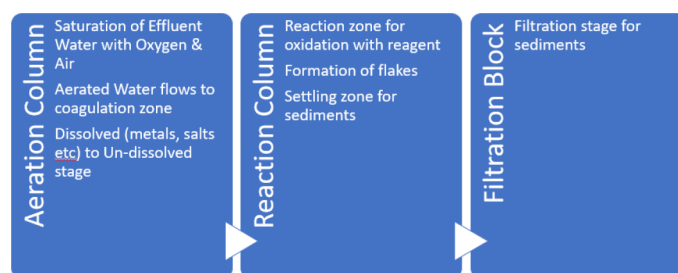


Figure 3B—Schematic For Process Treatment Stages

The system incorporates:

- **Aeration Column:** Ensures oxygen saturation and promotes initial mixing.
- **Reaction Column:** Facilitates oxidation and floc formation.
- **Settling Zone:** Enables gravity-based separation of precipitated solids.
- **Filtration Blocks:** Provide single or multistage particle removal to achieve final clarity.

This modular approach ensures that unstable oxidants are generated and applied on-site, improving reactivity and reducing storage hazards. Treated effluents are evaluated for residual contaminants before discharge.

Sensor-Based Monitoring Setup

The concentration of sodium ferrate is monitored by the sensor installed in the cell. The sensor is a photometric sodium ferrate concentration sensor using laser sources. The flow through sensor utilizes two spectral ranges for measuring the sodium ferrate concentrations.

Figure 4 shows the setup for the lab test conducted to test different contaminated streams in the laboratory. Setup can show the pumps, anode and the supply drain connections for understanding of the operation. Power controllers are also connected with the generator for the power supply.



Figure 4—Sodium Ferrate generator setup for lab test

Laboratory Test Data

Laboratory tests were conducted on four industrial wastewater samples from different types of industrial plants, measuring TDS, TOC, COD, $\text{NH}_3\text{-N}$, TSS, BOD, sulfate, sulfide, and sodium pre- and post-treatment as applicable. The study evaluated contaminant degradation by analyzing pollutant concentration reductions across different samples.

The samples were gathered from different diversified industrial areas to evaluate the diversification of the process application and the impact of sodium ferrate treatment for the wastewater treatment. Sample 1 is from fertilizer plant having contaminants in the wastewater stream. Sample 2 is from the petrochemical plant having high level of contamination as well. Sample 3 was collected from the paint and solvents industry having high contamination as well. Sample 4 was collected from the heavy industry petrochemical plant involved in production of different product as of sample 2.

Table 1—Fertilizer plant complex wastewater stream sample test results

Parameter	Inlet (mg/l)	Outlet (mg/l)
BOD	1300	22
TOC	23000	2000
COD	51450	5983
NH_3	20100	3640
Mo	7	0.03

Table 2—Petrochemical plant complex wastewater stream sample test results

Parameter	Inlet (mg/l)	Outlet (mg/l)
TDS	68834	16978
TOC	640	90
COD	200	315
TSS	40	68
Mo	<0.004	0.005

Table 3—Paint & Solvents plant complex wastewater stream sample test results

Parameter	Inlet (mg/l)	Outlet (mg/l)
TDS	182	196
TOC	190	130
COD	627	409
TSS	15	37
BOD	640	<2

Table 4—Heavy Industrial plant complex wastewater stream sample test results

Parameter	Inlet (mg/l)	Outlet (mg/l)
TDS	1406	2049
TOC	730	49
COD	2431	156
TSS	12	130
Mo	1.1	0.3

Observations, Results and Discussion

Laboratory tests were performed on wastewater samples from four industrial facilities, including fertilizer plants, petrochemical units, and paint and solvent manufacturers. Each sample underwent both pre- and post-treatment analysis. Parameters measured included Total Dissolved Solids (TDS), Total Organic Carbon (TOC), Chemical Oxygen Demand (COD), Biological Oxygen Demand (BOD), Total Suspended Solids (TSS), ammonia (NH₃-N), molybdenum (Mo), sulfates, and sulfides.

Sodium ferrate treatment was applied using the in-situ generation method described earlier. Reaction, sedimentation, and filtration processes were performed according to the 3-stage modular design. Each sample was tested under controlled conditions, and the results were tabulated to assess contaminant removal efficiency.

General Observations

The system consistently reduced a wide range of pollutants across all tested samples. Notably:

- **COD** reduction reached up to **88%**
- **TOC** reduction exceeded **95%**
- **BOD** reduction was over **99%** in several cases
- **Mo**, a heavy metal, showed **over 95% removal** in critical cases

- Some samples showed a minor increase in TSS, attributed to reaction-induced flocculation

Sodium ferrate effectively reacted with organic and inorganic contaminants, forming insoluble complexes and radicals that precipitated out and were filtered during downstream processing.

Sample-Specific Results and Insights

Sample 1 - Fertilizer Plant

High TOC, BOD, ammonia, and COD levels were significantly reduced. *BOD dropped from 1300 mg/L to 22 mg/L ($\approx 98.3\%$ reduction), and TOC from 23,000 to 2000 mg/L ($\approx 91.3\%$).* Molybdenum was reduced from 7 mg/L to 0.03 mg/L.

This performance confirms the ability of ferrate to act on both organic and nitrogen-based contaminants while precipitating heavy metals. The treatment required eight filtration passes and 100 mL of ferrate solution, suggesting that heavily loaded effluents may require higher dosing and filtration cycles to meet discharge limits.

Sample 2 - Petrochemical Plant

High TDS and COD were observed. *TDS dropped from 68,834 to 16,978 mg/L ($\approx 75.3\%$ reduction).* TOC and COD also improved, with filtration passes up to 3. Slight increase in TSS was observed, likely due to flake formation from ferrate reaction.

While TOC and TDS were reduced significantly, the increase in COD and TSS suggests the presence of oxidizable intermediates or unreacted ferrate residue. This reinforces the need for final-stage polishing, especially for petrochemical streams with complex hydrocarbons.

Sample 3 - Paint and Solvent Industry

COD reduced from 627 to 409 mg/L ($\approx 34.8\%$) and TOC from 190 to 130 mg/L ($\approx 31.6\%$). BOD dropped below detection limits (<2 mg/L). TDS increased slightly, suggesting partial ferrate carryover.

Despite excellent BOD removal, the system showed moderate efficiency for TOC and COD. This may reflect partial oxidation of solvents and paint residues into soluble by-products, underscoring the need to optimize dosage and contact time for this wastewater type.

Sample 4 - Heavy Industrial Petrochemical Plant

TOC reduced from 730 to 49 mg/L ($\approx 93.3\%$), COD from 2431 to 156 mg/L ($\approx 93.6\%$), Mo from 1.1 to 0.3 mg/L. The sample had high inorganic loads; filtration was critical for performance.

This effluent was less organically loaded than Sample 1 but had significant inorganic content. The increase in TDS and TSS is attributed to reaction-induced precipitation and minor carryover of ferrate. These findings validate the system's heavy metal removal performance while also identifying areas for enhanced solid-liquid separation.

Results Interpretation and Operational Learning

The process demonstrates a high degree of adaptability and efficacy in removing a diverse spectrum of contaminants. Ferrate's high redox potential enables degradation of recalcitrant pollutants, including hydrocarbons, ammonia, and heavy metals, through oxidative flocculation. The formation of flocculent by-products was common across all reactions, aiding sedimentation but contributing to increased TSS if not fully filtered.

Minor increases in TSS and TDS post-treatment are attributed to:

- Reaction by-products (oxidized sediment flakes)

- Unreacted ferrate residues in insufficiently filtered samples

These can be mitigated by optimizing dosage, extending filtration stages or improving filter media, as applicable.

Filtration cycles (3 to 8 passes) played a key role in achieving high clarity and compliance. Optimizing these steps for different wastewater types is crucial for industrial adoption. Controlled dosing, real-time sensor-based regulation, and tailored reactor sizing contributed to reliable performance across diverse water chemistries.

Implications for Industrial Applications

On-site ferrate generation provides several advantages:

- Avoids bulk chemical handling
- Reduces CAPEX and OPEX
- Scales easily with parallel modular units
- Enables compliance with strict effluent regulations without major infrastructure changes

The 3D-printed system offers scope for future scaling through automation and remote monitoring. The system's adaptability to in-situ operation, combined with reduced chemical handling and compatibility with renewable power, supports its role in sustainable wastewater management. Its ability to achieve >90% contaminant reduction in four different industry samples highlights the robustness and universality of the solution.

Conclusion

This study demonstrates that in-situ electrochemical generation of sodium ferrate is a promising, adaptable solution for treating complex industrial wastewater. Across a diverse set of high-contaminant effluents, the system achieved significant reductions in BOD, COD, TOC, ammonia, and heavy metals—validating both its oxidative efficacy and its scalability through a modular setup.

While results confirm the system's robustness and operational flexibility, further research is warranted to:

- Optimize alkali solutions and electrode configurations for enhanced ferrate yield;
- Refine real-time dosage control mechanisms to minimize reagent waste and TSS/TDS carryover;
- Tailor treatment sequences for specific wastewater chemistries, especially those involving solvents or hydrocarbon derivatives.

Continued evaluation of reaction dynamics, material durability, and energy input parameters will support future scale-up and integration into industrial wastewater infrastructures. The findings underscore sodium ferrate's potential as a sustainable, low-footprint, and high-performance oxidant for on-site water treatment applications.

References

1. Alsheyab, M., Jiang, J. Q., & Stanford, C. (2009). On-line production of ferrate with an electrochemical method and its potential application for wastewater treatment: A review. *Journal of Environmental Management*, **90**(3), 1350–1356.

2. Andreev, S. Y., et al (2020). Investigation of the technological process of electrochemical synthesis of sodium ferrate in anode cells of a membrane electrolyzer. *Regional Architecture and Construction*, **2**, 142–149.
3. Talaiekhosani, A., Talaei, M. R., & Rezaia, S. (2017). An overview on production and application of ferrate(VI) for chemical oxidation, coagulation, and disinfection of water and wastewater. *Journal of Environmental Chemical Engineering*, **5**(2), 1828–1842.
4. Golovko, D. A. (2013). Features of electrochemical production of sodium ferrate (VI) solutions from secondary raw materials. *Bulletin of the National Technical University Kharkiv Polytechnic Institute*, **47**, 26–33.
5. Petkova, A. P., et al (2025). Sodium ferrate-based oxidation for wastewater treatment and reservoir reuse. *IJE Transactions B: Applications*, **39**(2), 523–533.
6. Koniashina, A., et al (2024). Wastewater treatment and preparation for reservoir pressure maintenance using sodium ferrate solution produced on-site by membrane electrolysis. Paper presented at the SPE Conference at Oman Petroleum & Energy Show, Muscat, Oman, April 2024. DOI: <https://doi.org/10.2118/218566-MS>